



CS 310: Introduction to Communicating Distributed Processes

Course Code	CS 310
Course Name	Introduction to Communicating Distributed Processes
L-T-P-C	3-0-0-3
Prerequisites	CS 307 Systems Practicum
Intended for	UG
Distribution	Discipline core for CSE Semester: 6th semester
One-time approval	recommended by the 9th BoA (Oct 2015)
Revised	15th BoA (Dec 2016) and 12th Senate (Mar 2017) - credits changed to 3-0-2-4

Course Contents

- **Introduction:** Operating systems, Network Architecture
- **Process management** Scheduling, Inter-process communication
- **Network architecture:** Socket address, layering, encapsulation
- **Packet Switching:** Routing, switching, addressing
- **Process coordination:** Scheduling, Synchronization, deadlocks
- **Network reliability:** Protocols, TCP/UDP, Flow Control, Congestion Control
- **Network and distributed operating systems:** Synchronization, mutual exclusion, concurrency

Text Books

1. A. Silberschatz et al, **Operating System Concepts**
2. A.S. Tanenbaum, **Computer Networks**

References

1. A.S. Tanenbaum, **Modern Operating Systems**, 3rd Edition, Pearson Ed., 2008
2. B. Davie & L. Peterson, **Computer networks**, 4th Edition, Morgan and Kauffman, 2011.
3. B. Forouzan, **Introduction to Data Communications and Networking**